

Element H

Testing Plan:

Design Requirement	Tests
Changes color manually and automatically	The group will measure this by seeing how many color options are available on the board. The group will measure the rate at which the lights change by calculating the rate of flashing of the lights. The group will measure the input > output speed in order to see how quickly the colors can be changed manually.
Provides calming effect	The group will test this requirement by measuring the speed at which the lights change for different pattern options on the board. Different speeds provide different effects, so knowing the different ones will be useful for the teachers when choosing an option. The group will measure this by having the teachers record the behavior of the children before and after being exposed to the board. The group will also have the teachers measure how much time the kids spend in front of the board in order to discover if they enjoy it or not.
Illuminates the sensory room	This will be tested by measuring how far the light from the board reaches in the sensory room. The group will use a luminosity sensor to measure how far the light reaches on the board's highest brightness and lowest brightness.
Brightness can be altered	This will be measured by calculating the brightness in Lux at the lowest and highest settings, as well as counting the amount of brightness settings.
Integrated into the room	This will be a visual test of how well the board fits into the cohesiveness of the room. The

	group will determine this after installing the board by judging how well it fits into the room. Additionally, the group will ask the teachers and children to give their opinions on how well the board fits in, and if it gets in the way at any points.
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Use of Field Experts:

Dr. Sarver recommended that the testing plan should include a measurement of the light produced by the board, and he recommended utilizing an app called LightMeter LM-3000. The group incorporated this into the testing plan.

Dr. Dougherty recommended that the testing plan should include a measurement of the speed at which the lights change during each of the boards preset patterns. The group incorporated this into the testing plan